

## 9 Sustaining mobile learning with pervasive game

An example of cultural  
history exploration

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### Introduction

In recent years, the advancement of digital technology has changed the practice of human society. The advancement of computer technology has attracted intensive and crucial discussions and debates on the issues of “learning to use technology” and “learning with technology” in school education. In the realm of communications technology, digital convergence has integrated the web, digital platform, Web 2.0 and cloud technology, have turned the one-way transmissive mode of teaching to collaborative knowledge construction and sharing. Advancements in information technology beyond the year 2000 have pushed the instructional applications of computer technology to go further to break through the limitations of time and space to extend the classroom beyond its physical constraints. The innovation of mobile and interactive technology has made communication in the ubiquitous context possible. The terms “mobile learning (mlearning)” and “ubiquitous learning (u-learning)” have become more well-known to the general public. The mobile device is subject to no constraints in time and space. Laine et al. (2010) defined mlearning as a form of informal learning where the learner traverses a physical context or contexts carrying a personal mobile device which provides learning materials and activities. Mikic et al. (2007) believed that mobile learning has created a new learning environment that enables learners to learn at any time and anywhere, through action learning aids, teaching materials, teachers and other learners.

The 2009 Horizon Report has predicted mobile learning to be an important development for digital learning (Johnson et al., 2009) and the 2011 Horizon Report predicted that more than 80% of people will use mobile devices connected to the Internet in 2015. More importantly, by 2016, mobile device use in education would outnumber the use of computers (Johnson et al., 2011) in the classroom. With the help of advanced mobile devices and well-designed instructional strategies, mlearning is enabling students to walk out of classrooms and into real life scenarios to interact

with real learning environments. Studies (e.g. Laine et al., 2010) have also showed that mlearning and u-learning are especially helpful in facilitating learners' construction of knowledge in regard to the situational information in social, cultural and physical contexts.

Well-designed instructional strategies are essential for successful mlearning. However, many local studies on mlearning simply reiterated the concerns of establishing a learning system and emphasising the use of the mobile devices (such as PDA) with the established system (e.g. Liang, 2005; Lin, 2006). These studies appear to fall short of designing adequate strategies to carry out effective instruction. The system-design type of studies may have caused instructors, especially high school teachers, to be reluctant to adopt mobile technologies due to the lack of budget support and knowledge of applicable instructional strategies. Therefore, Ng and Nicholas (2013) proposed a person-centred model indicating "good communication and support, being consultative and maintaining trust between management, teachers, students and technical support personnel" (p. 4) as the core actions for implementing mobile learning.

Other than the instructional issues, making mobile learning sustainable is the current issue. Cisler (n.d.) discussed the sustainability issue of information and communication technology (ICT) in education that encompassed four dimensions, namely economic sustainability, social sustainability, political sustainability, and technological sustainability. It became important to seek a mechanism that would eliminate technological implementation costs, and intrigue users to choose knowledge inquiries as individuals and as groups. As a result, in this research, a pervasive game was used to design mobile learning using participants' self-owned mobile phones with Internet connections, their personal social network service accounts and leisure time; therefore little technical and instructional efforts were required to implement this inquiry-based learning strategy.

An innovative way to accomplish mlearning using the instructional pervasive gaming (IPG) approach has been introduced by various educational researchers in the past ten years (Laine et al., 2010). Montola (2011) explained a pervasive gaming environment as "an extension of mlearning with an emphasis on the roles of an intelligent environment and of the context". The advancement of mobile technology has brought about not only the booming of m-business but also the imaginary space of mobile games (Chen, Guo, & Shih, 2012). Okazaki, Skapa, and Grande (2008) recognised that mobile games have quickly opened up new dimensions for entertainment applications and have rapidly become an attractive alternative to PC-based games.

In mobile learning strategies, game-based mobile learning has been recognised as effective (Huizenga et al., 2009; Schwabe & Göth, 2005). Related research (Facer et al., 2004) also demonstrated that pervasive gaming has significant effects on learning. Recent research (Schwabe & Göth, 2005; Shih & Chen, 2012) indicated that mobile gaming has positive user effects,

especially in educational applications. Schwabe and Göth (2005) designed an instructional system that combined mobile learning and mobile gaming. This instruction used college students as learners who were divided into groups to play the game. They obtained information from the website regarding the mission and geographical coordinates of the locations. They then used GPS to complete a task and uploaded the result back to the workstation before they could get the location information of the next mission. The winner was the group who completed the task first.

In mlearning, the learning of content is location-sensitive as learners are free to move around. The advantage of employing mobile technology in learning is not the mobile technology itself, but the unique attributes of the mobile devices that provide an environment upon which one can apply innovative learning theories and learning strategies (Su, 2005). With the aid of GPS, Chen and Shih (2011a; 2011b) initiated a study employing a typical pervasive game, geocaching, to teach high school students geographical coordinates and map reading. The results indicated that GPS-aided geocaching helped students to learn to read maps and recognise the geographical coordinates in a more effective way. In addition, the GPS-aided geocaching group generally acquired a better attitude toward the use of technology.

Such results are in accordance with some other related studies (Chavez, Schneider, & Powell, 2004; Mandy, 2006). However, Gentes, Guyot-Mbodji and Demeure (2010) pointed out that most game strategy does not take into consideration the anthropological data pertaining to the specific context of use. They argued that the advancement of geolocalisation helps us to concentrate on the diversity of location-sensitive experiences rather than focusing uniquely on the geographical skills of players engaged in a treasure hunt pursued by other players in their game design. They further indicated that maps and trajectories are one aspect of treasure hunts, while Shih and Chen (2012) indicated that companionship and discovery are at the core of the pleasure of pervasive games. Walther (2005) recognised that a mobile game is one that takes into account the changing of relative or absolute position/location in the game rules. This excludes games for which mobile devices merely provide a delivery channel where key features of mobility are not relevant to the game mechanics. Corey (2010) stated that both urban and environmental histories in particular lend themselves to hands-on approaches that link the classroom with tangible examples of historical events and trends through the pedagogical design of the learning of place. Huizenga et al. (2009) also claimed that mobile and location-based technologies provided opportunities to embed learning in authentic environments, thereby promoting enhanced engagement and learning outside of traditional formal educational settings. They conducted a quasi-experiment where the students from 10 classes played a mobile history game while the students from another 10 classes received a regular, project-based lesson. The results showed that those students who played the game

were more engaged and gained significantly more knowledge about medieval Amsterdam than those students who received regular project-based instruction.

The goal of this research was to establish a pervasive game framework to design an exploratory game that allows participants to explore historical locations to learn about interrelated knowledge. The historic site, Provintia, in Tainan, was chosen to be the gaming location. Its related history and cultural content were embedded in the multimedia form with augmented reality (AR) interactions. The participants used the social networking service, *Facebook*, to document group activity, share information, and interact with group members. The implications for sustaining pervasive gaming were explored and discussed.

## Literature review

### *Pervasive games*

A pervasive game (PG) is a mobile game that seamlessly integrates the physical environment with virtual scenarios. According to Walther (2005), a PG implies a mixed-reality gaming space; that is, the gameplay connects the virtual world with the tangible world. He further indicated that the mobile mechanisms (that include software, hardware and information), should be in place throughout the gameplay to facilitate and ensure that there are fluent communication channels between the virtual and physical worlds. Pervasive games often blur the boundary between games and real life.

Laine et al. (2010) commented that PG is an extension of mlearning. The interaction between the player and the environment is the key point. Another key point about PG was raised by Montola (2011). He stated that the “magic circle of play” created by an online game is an invisible fence keeping the game from real life. He further explained that the magic circle of play is a social and cultural contract that separates ordinary life from play, where players are also separated. However, PG breaks the circle and bridges real and virtual environments to accelerate more interactions.

From an educational view point, Laine et al. (2010) described a technology integration model for game-based pervasive learning systems, stating that an instructionally constructed PG needs to meet the following requirements: (1) pedagogical requirements that include user profile, interaction and collaboration, ownership, authenticity and relevance, as well as provision of support and assessment; (2) game design requirements that consider resources (both financial and human), cultural issues, technical issues, environmental issues, social issues and temporal issues; and (3) context requirements that include context-awareness, dynamics, interaction and content.

There are various papers that discuss the design issues of pervasive games. From the perspective of technology applications, Walther (2005)

constructed a model and identified the “four axes of PGs”. He illustrated the four axes as: (1) distribution: this refers to the effective wide distribution network of the gaming information; (2) mobility: this refers to user mobility, computing mobility, network mobility, and context-aware and cross-platform services; (3) persistence: this refers to the total availability all along; and (4) transmediality: this refers to a media circle that multi-links the world of virtual social networks. He further depicted three critical constructional framework elements of PG: (1) game rules: all PGs must be rule-based and clearly defined; (2) game entities: the abstract class of an object that can be moved and drawn over a game map, which can further be shaped into three categories such as game object, human agent, and physical object; (3) game mechanics: an input-output engine that monitors and modifies the physical and virtual linkage to ensure a fluent game flow.

In PG, players are typically equipped with handheld or mobile devices. WiFi or 3G connections are common technologies used for the players to communicate with the virtual environment. To be more precise, location tracking and orientation sensors are usually based on GPS. AR is also used to increase the gamification of gameplay. According to Benford et al. (2005) pervasive games need three kinds of technology support: (1) displays that can make digital content, which can transmit digital content to users’ mobile devices, such as smartphones, PDAs, and tablet PCs; (2) wireless communications, which can connect users with the server and other users by 3G, WiFi, GPRS, and GSM; (3) sensing technologies, which allow the context-awareness mechanism to work for the environment, such as GPS, microphone, camera, other sensors. These mobile technologies allow users to experience the game activities outdoors.

As mentioned above, Gentes, Guyot-Mbodji and Demeure (2010) specifically pointed out that recognition of local culture must accompany the co-creation of content and team exploration. It also strengthens the importance of cooperative learning in pervasive game. According to Pragnell, Roselli and Rossano (2006), cooperative learning is achieved by dividing learners into small groups to accomplish the best results by means of mutual assistance among the group members. This fosters both a team spirit and interpersonal relationship with positive personal effectiveness.

There are many collaborative and cooperative learning studies, but only a few have been discussed in the gaming context. Dyson, Griffin and Hastie (2004) indicated that students can compete more effectively working as a team against another team in a game unit, while Huang, Shih and Lai (2011) studied a cooperative game used to increase the learning effectiveness of an engineering class, and the result was that students in the cooperative learning class significantly outperformed students in other classes. Students’ feedback also showed that cooperative learning effectively increased their motivation to learn. The results of these collaborative or cooperative game studies support Gentes, Guyot-Mbodji and Demeure’s

(2010) and Edwards' (2003) claim that team cooperation, collaboration, and competition are essential for gameplay.

### *Augmented Reality*

Augmented Reality (AR), also known as Extended Reality, is a computer graphics interactive technology originating from Virtual Reality (VR). Azuma (1997) believed that interactive technology could be applied to military, medicine, industrial design, maintenance, commercial activity, learning and entertainment. Therefore, in recent times, AR has often been used to enhance interaction in the pervasive game and mobile learning activities. So far, AR content recognition techniques include three main types: (1) use of specific images as recognition objects; (2) use of real objects as image recognition targets; (3) use of GPS positioning and digital compass to identify images. As Ultra Mobile PC (UMPC) image processing technique has improved, more and more studies have implemented AR into their mobile learning practices.

AR seems to be essential to pervasive games or mobile edutainment applications. For instance, Cheok et al. (2004) brought the classical arcade game *Pacman* into the real world and called it *Human Pacman*. Players collected cookies made by AR and avoided being caught by the ghosts. The ghosts in the game are avatars representing other players, so it is also a multiplayer game. Other games, *ARQuake* (Piekarski & Thomas, 2002) and *Treasure* (Barkhuus et al., 2005), also used the application of AR to allow the players to collect objects.

Herbst et al. (2008) presented an interactive time travel game, *TimeWarp*, which guides the users to explore historic buildings with the AR virtual tour guide, *Heinzelmannchen*. Participants explored the two-thousand-year history of the city of Cologne, and experienced Medieval Roman times, the new century, and even future time travel. AR in this instance functions not only as a simple game tool, but also as the mobile guide. Ballagas, Kuntze and Walz (2008) designed *REXplorer* as a historical exploration guide for tourists in Regensburg. Tourists visited landmarks and found virtual magical spirits and treasures using detectors on their way.

Billingshurst et al. (2003) claimed that AR is useful to education because using AR as teaching materials can allow students to be immersed in the learning of content of both the virtual and real environment. Students can even succeed in such learning environments without computer experience. In the past decade, the increased simplicity of using AR has brought students closer to the virtual scene as well as the real environment. Perceptual feedback creates fun learning and stimulates students' learning motivation and enthusiasm. AR was used in several educational situations, such as geometry teaching, science experiments (Newton's Law), and nuclear emergency practice (Enyedy et al., 2012; Kaufmann, 2006). The research results suggest that AR can increase students' understanding in the learning of content.

### *Social networking services*

Traditional digital learning (e-learning) has mostly taken the form of using a learning management system. Course content was totally decided by the instructors. Following the development of Web 2.0, the control rights of digital learning have been transferred to the students. A typical Web 2.0 tool is *Wikipedia*, in which people edit the digital content together and learn independently through the process (O'Hear, 2006). Today, the most used Web 2.0 application is the social networking service *Facebook*. Instead of using social networking as a learning platform Liu (2010) used it as a teaching tool to increase students' learning effectiveness.

Mazman and Usluel (2010) believed that social networking services could be used as a teaching channel to extend classroom learning. At the same time, students can interact, discuss, and share information on it, which further helps to develop critical thinking ability. The results of their study suggested that *Facebook* could be effectively used in education. OnlineCollege.org (2009) listed 100 ways of *Facebook* teaching, illustrating the benefits *Facebook* can bring to sustain group communication. In pervasive gaming research, *Facebook* is used for group cooperation, information management, and sharing.

### **Research methods**

The research process was divided into four stages (see Figure 9.1). The first stage was the learning content preparation. The game script was designed around Tainan historical events that happened in the period of Koxinga, and also contained some folklore stories to increase the fun. The activity space was a Grade I national historic monument, Provintia. Provintia was built by the Dutch in 1653 as a trade and administrative centre. Today, the buildings are in the Chinese architectural style of the Ching dynasty because the original fort was damaged a long time ago and became the building foundation of today's Wen Chang pavilion and Temple of Sea God. These buildings have been through times of Dutch, Kingdom of Tungning, Ching Dynasty and Japanese colonial rule, until the Taiwanese restoration. It is the epitome of history with various cultural artefacts and stories.

The learning content of the game was collected through historical literature, books, textbooks, web information, Tainan government tourist bureau information, field study, tour guides, and oral history interviews. The learning content includes four aspects: (1) a historical aspect that includes stories about the fort, ordnance depot, hospitals, and schools in various historical times; (2) a geographical aspect that is largely about the importance of the location and position of the fort during wartime; (3) an architectural aspect that includes objects and features such as stone lions and sculptures; and (4) a cultural aspect that includes historical monuments such as tombstones and wells. In the pervasive game, 14 riddles were

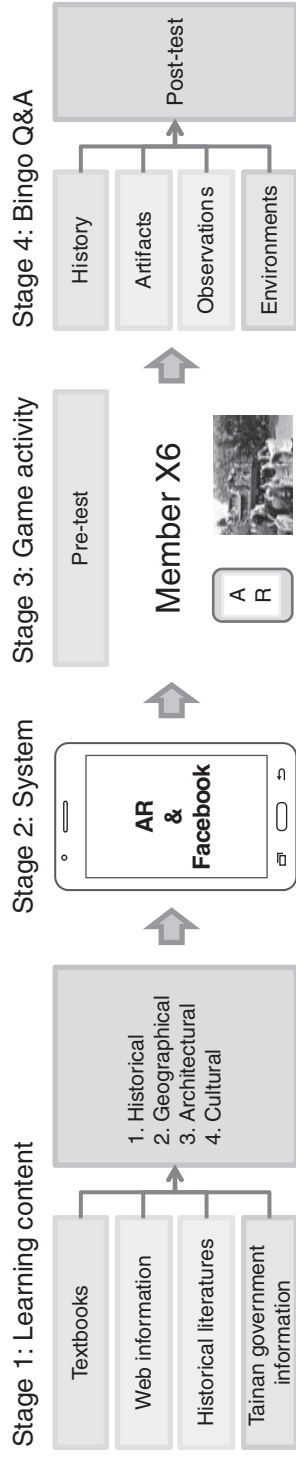


Figure 9.1 The research process



created for the exploration. The same questions were used in both the pre-test and post-test of learning effectiveness.

The second stage was system design. Smartphones equipped with wireless Internet connections were used as the mobile learning tools. The phones were installed with AURASMA (Figure 9.2). AR presentations on the players' mobile phones can increase players' gaming motivations and interests. With the instructional pervasive game design, players can have more interactions with the environment. *Facebook* was installed to allow participants to communicate and share answers with group members. It was not only a communication bridge to players, but also a terminal for researchers where they could oversee the participants' interactions and monitor players' gaming conditions. The game mechanism was implemented to increase group competition and cooperation.

The third stage was the pervasive game activity (Figure 9.3). Thirty college students were invited to participate in the study. They were divided into groups of five with a total of six groups. Every group was accompanied by one research observer whose responsibility was to offer task guidance and solve hardware or software problems. Before the game, *Facebook* group spaces were installed for each group. Then, a pre-test was conducted to gather data on the players' prior knowledge about history. At the same time, a pervasive game effectiveness evaluation pre-test was also conducted to gather information on participants' expectations of the game.

When the participants entered Provintia, tourist brochures were distributed. The pervasive game started with riddle solving. Participants explored the historical spots or cultural objects according to the hints given by the riddles, trying to understand their contents and meanings. The participants used smartphones or tablet PCs to detect the target spots or objects and drive

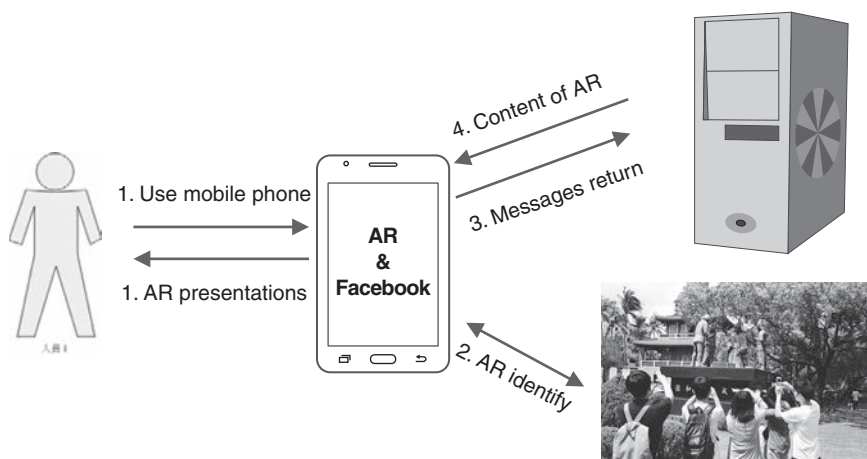


Figure 9.2 Aurasma system data flow

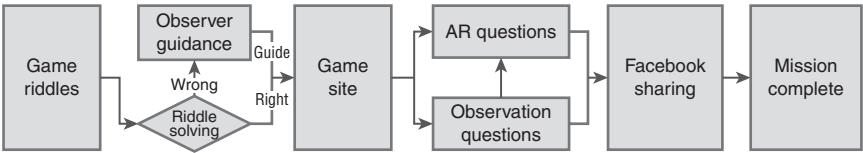


Figure 9.3 Pervasive game process



Figure 9.4 AR content

the AR content to show short clips of historical events or talking figures on the wall to give hints (Figure 9.4). The AR content would repeat until the participants had found the answers to the riddles, documented their findings, and uploaded photos or answers to the *Facebook* community.

After the pervasive game exploration, a group bingo competition of Q&A activity was conducted. The groups took turns to answer questions. There were 49 questions concerning history, artefacts, observations, and environments. They were allowed to retrieve information or answers from the Facebook communications. The purpose of this activity was to get all of the participants together, and refresh their memories of the

one-day activity. It was also important for the instructor to call a close to the activity.

The final stage was the data analysis. In the gaming process, all interactions were recorded. All learning paths and the learning content entered during the process were recorded on the mobile devices. Research observers also recorded behaviour observations. After the game, historical knowledge post-tests were conducted. A pervasive game evaluation questionnaire and focus group interviews were also conducted to investigate the effectiveness of the game. They included four aspects, namely game rules, game mechanism, game interaction, and game content. Behaviour observations and interaction records were used for cross-analysis of the evaluations at the end.

## Research results

In terms of learning effectiveness, the results of the pre-test ( $m=31.33$ ) and post-test ( $m=88.50$ ) showed that the participants' understanding of Provincia history and culture had significant improvements ( $t=22.46$ ,  $p=.000<.001$ ) from the pervasive game (Table 9.1).

Upon analysis of the four parts of the questions, it was found that participants performed better on geographical aspect and architectural aspects, 93.3% and 92.3% respectively on the correction rates; and less well on the historical aspect and cultural aspect, 89.4% and 82.7% respectively on the correction rates. From the student interviews, out of the four types of questions, e.g. history, artefacts, observations, and environments, it was deduced that students performed better on the observational questions because the learning content was more tangible and the related concepts were easier to grasp. Even though the historical and cultural content were more difficult since they were more conceptual and abstract, the AR content provided the participants with cognitive support since the short clips would provide clues to the users. At the same time, it had stimulated students' learning motivations to continue the exploration.

In terms of the pervasive game's effectiveness, four aspects including game rules, game mechanism, game interaction, and game content, were evaluated. Since the pervasive game's effectiveness evaluation pre-test was conducted to gather information on participants' expectations of the game, the post-test was conducted to show how the pervasive game compared to

Table 9.1 T-test results of learning effectiveness of pervasive game

| Item                                        | pre- and post-test | n  | mean  | SD    | t      |
|---------------------------------------------|--------------------|----|-------|-------|--------|
| Provincia history and culture understanding | Pre-test           | 30 | 31.33 | 12.65 | 22.46* |
|                                             | Post-test          | 30 | 88.50 | 7.44  |        |

\* $p < .001$

Table 9.2 T-test results of pervasive game activity effectiveness

| Item             | pre- and post-test | n  | mean  | SD   | t       |
|------------------|--------------------|----|-------|------|---------|
| Game Rules       | Pre-test           | 30 | 25.36 | 2.85 | -.803   |
|                  | Post-test          | 30 | 25.86 | 2.87 |         |
| Game Mechanism   | Pre-test           | 30 | 23.73 | 3.00 | .22     |
|                  | Post-test          | 30 | 23.56 | 3.21 |         |
| Game Interaction | Pre-test           | 30 | 29.30 | 3.09 | .100    |
|                  | Post-test          | 30 | 29.23 | 3.43 |         |
| Game Content     | Pre-test           | 30 | 12.90 | 3.97 | -9.950* |
|                  | Post-test          | 30 | 20.96 | 2.35 |         |

\*p < .001

the participants' expectations. The t-test results (Table 9.2) show that the pervasive game design was very close to the participants' expectations although some of the results between pre- and post-tests did not reach significant differences.

The game rule aspect investigated whether the challenges, competitions, and feedback would influence the participants' gaming behaviours and attitudes. It included five questions such as "I thought the game rules were clear"; "The bingo game had stimulated me to be more actively involved in the riddle solving process"; and "Solving riddles had given me a great sense of achievement". The pre-test (m=25.36) and post-test (m=25.86) had no significant difference (t=-.803, p=.429>.05). It showed that the game was about the same as the participants' expectation. The participants said in the interviews that they felt excited and happy when able to solve the riddles, which also enhanced their sense of achievement.

The game mechanism aspect investigated whether the use of smartphone, AR, and *Facebook* would influence the participants' gaming behaviours and attitudes. It included five questions such as "AR storytelling had made the site exploration more interesting"; "AR storytelling made me have deeper impression to the historical site"; and "I would read the materials on the *Facebook* posted by other members to gain more understanding to the historical site". The pre-test (m=23.73) and post-test (m=23.56) showed no significant difference (t=.228, p=.821>.05), and the post-test result was lower than the pre-test. From the interviews, participants revealed that they still preferred shorthand note-taking instead of typing into the smartphone so the game mechanism was not as convenient as they expected. Therefore, the contents on *Facebook* were mostly photos or short notes. To improve this in the future, it might be useful to design a quick upload mechanism, keywords, or icons to enhance the participants' motivation in using *Facebook*. Nevertheless, some groups used the *Facebook* communication records to correctly answer the bingo questions at the end of the pervasive game. It showed that *Facebook* usage was quite helpful in some way.

The game interaction aspect investigated the participants' interactions with others and the environment. It included six questions such as "I found information that was not stated on the tourist guide"; "I would cooperate and share works with group members during the gaming process"; and "I like the AR contents more than the textual descriptions on the tourist signs". The pre-test ( $m=29.30$ ) and post-test ( $m=29.23$ ) had no significant difference ( $t=.100$ ,  $p=.921>.05$ ), and the post-test result was slightly lower than the pre-test. From the interviews, participants pointed out the hardware factors had influenced their motivation to continue the game with enthusiasm. Sometimes AR image recognition failed due to sunlight, shadows, or photo taking angles. Therefore, a rich image database might be necessary to increase the interaction fluency.

The game content aspect investigated the participants' meta-cognition of the game. It included four questions such as "I have more understanding of the historical background of the site"; "I have more understanding of the architectural features of the site"; and "I have more understanding of the cultural artefacts of the site". The pre-test ( $m=12.90$ ) and post-test ( $m=20.96$ ) had reached significant difference ( $t=-9.950$ ,  $p=.000<.001$ ). After playing the pervasive game, participants generally had more understanding of Provincia, and could even tell the stories to others.

## **An IPG framework**

PG is a relatively new term for most instructional gaming researchers and developers. The idea of PG started at the beginning of this century. Even though numerous mixed-reality game prototypes have been built and studied (Montola, 2011), a comprehensive model and guidelines for developing instructional PG are still being developed.

Gentes, Guyot-Mbodji and Demeure (2010), recognised local culture to be a part of pervasive gaming, especially with Instructional Pervasive Game (IPG). They suggested that for a cultural-rich PG, three key features should be included: (1) collaborative content where the content should be designed by the local community – people with qualified knowledge or who are actually living on the premises, therefore, scholars, neighbours, and respectable people alike are all contributors of the content; (2) team exploration where the gameplay either relies on solitary errands or on collective sharing involving strategies that can be more "group oriented"; and (3) cultural narrative where there is always a first narrative that sufficiently describes the culture of the city to provide game players with a clear view of the fashionable places, the living or working areas, and the cultural spots.

A review of relevant literature and IPG implementation experiences helped in the construction of a theoretical framework. The framework defines that a cultural-inquiry PG should accommodate technology convenience, cultural richness, and game fluency. A set of criteria was also defined to verify the game flow, the game roles and contexts, and game collaboration. A conceptual

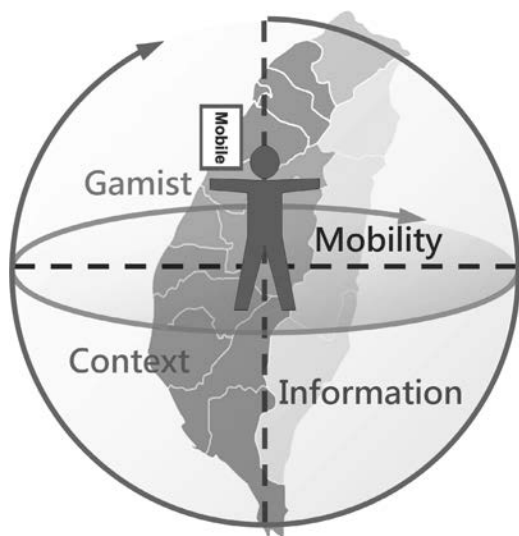


Figure 9.5 Instructional Pervasive Game learning model

model was formed in relation to the framework (Chen & Shih, 2012) (see Figure 9.5). The six factors of IPG included the user, environment, information, context, mobility and gamist.

The first two factors, user and environment, are the tangible factors in the model. The user is the student who is the central learning subject. Different from lecture-type instruction where the teacher is the centre of the learning process, students play the main role in the learning process in IPG. The learning environment refers to the physical environment, which is not confined to the classroom but the whole living environment where the students can physically reach and experience learning. In our case, Taiwan is the main location where learning happens.

The third and the fourth factors are the information and the context. The information refers to the teaching materials that are selected, designed, and organised by content experts (in most cases these are the teachers) which are taught both in class, presented on the mobile devices, and referred to in the physical environment and explored by the students. As the information given by the teachers seems to be linearly structured material, it guides the students to extend their learning spaces to the environment that forms a bigger space that is the learning context. As shown in the figure, the information axis leads the students to “spin” around the learning space that forms the surface of context. The learning context is the abstract and conceptual part of the learning, which acts as the horizon of the learning ground that is expanded into a surface. It includes both the mechanical data and the anthropological data that are experienced, collected, and

formed in the exploration process. It is also the process when the students are constructing their own knowledge. What's important is that the anthropological data are those qualitative raw materials collected by the students which include photos, interviews, field notes, and so forth. In other words, these two factors play the role of content and context.

The fifth and sixth factors are the mobility and gamist. Mobility refers to user's ability to move around in the learning environment. Other than sitting in the classroom, users not only have the freedom but also the capability to travel and explore with learning content embedded in the mobile devices. As the mobility is the horizontal axis that extends users' learning experiences to the outside of confined classrooms, gamist is the drive that enables the knowledge surface to expand. Gamist, the game rules that lead to cooperation or competition, is the driving force that creates the challenge and adventure scenario. Using a metaphorical description, if mobility is the vehicle that takes the users to places, gamist is the fuel that provides the energy and motivation to surf. The two factors create the surface consisting of motion and motivation.

The six factors together form the learning sphere. The whole sphere can be regarded as the knowledge body that is and is to be gained by the users. The IPG model reflects the Game Flow Model initiated by Sweetser and Wyeth (2005) that concentration, challenge, player skills, control, clear goals, feedback, immersion, and social interaction are key factors that may affect the effectiveness of a computer game. With the addition of the Pervasive Game Flow Model proposed by Jegers (2009), we can show 10 additional key factors for a PG are important to be integrated when implementing IPG:

- Support the players in the process of switching concentration between in-game tasks and surrounding factors of importance.
- Stimulate and support the players in their own creation of game scenarios and pacing.
- Help the players in keeping a balance in the creation of paths and developments in the game world.
- Be very flexible and enable the players' skills to be developed at a pace set by the players.
- Enable the players to pick up gameplay easily.
- Support the players in forming and communicating their own intermediate goals.
- Not imply or require players' actions that might result in a violation of social norms.
- Enable the players to shift focus between the virtual and the physical parts of the game world.
- Enable meaningful and purposeful social interaction within the gaming system.
- Motivate the players to communicate and interact socially.

## Conclusion

Since the early 1990s, active or inquiry-based education has received a great deal of attention. This active learning method has opened a new way for social science learning. The National Council for the Social Studies (1994) believed that students should “systematically employ processes of critical historical inquiry to reconstruct and reinterpret the past, such as using a variety of sources and checking their credibility, validating and weighing evidence for claims, and searching for causality” (p. 40). However, Molebash (2004) found that unguided online historical inquiry did not guarantee meaningful learning. Appropriate guidance is necessary for successful historical inquiry-based learning. Manlove, Lazonder, and de Jong (2009) asserted that supporting tools that incorporated goal-lists, hints, prompts, cues and templates reinforced the cognitive regulation of skills of students during a fluid dynamics task. According to their assertions, it could be reasonably assumed that their innovative pervasive game is an appropriate “supporting tool” for benefiting players’ information acquisition.

The goal of this research is to establish a pervasive game framework to design an exploratory game that allows participants to explore historical locations to learn about interrelated knowledge. From the trial of this study, suggestions about the use of a digital system would include the implementation mechanism of AR and *Facebook*. The use of markless AR can create a seamless learning environment combining virtual and real environments, and the use of *Facebook* can create group interaction and cooperation. Whether the t-test results between pre-test and post-test had reached significant differences, the scores between tests were close to each other. It meant that the effectiveness of pervasive game was close to the participants’ expectations. It showed that pervasive game not only can effectively help the participants understand the history, geography, and culture of Provintia, but also that it can allow the immersive learning mechanism to increase participants’ learning motivation.

The main theme of this chapter is to search for a sustainable model of mobile learning that doesn’t require high-level technical knowledge. Since smartphones are prevalent, related applications (apps) are also easy and free to download, and mobile Internet connections are common, mobile devices are indeed more generally used for learning purposes. For the purpose of making pervasive game widespread and knowing its sustainability, reaching participants besides youngsters who might not be as familiar with digital technology might be a concern. Once the mechanism is built, a city tour game or countrywide activity might also be applicable.

As this study came to a conclusion, Google Niantic Labs (2013) published a free Android App game, *Ingress*, which is a real-time multiplayer location-based game. Users check into places using the GPS function of their smartphone and interact with the portals established around public



arts or landmarks. Viewing alternate reality generated by AR technology, users participate in the making of an open narrative of science fiction. Countless users logged in to the game, which manifested the milestone of the transformation of traditional individual-based inquiry-based mobile learning into an interactive cooperative pervasive game. It also evidences the potential of what was discussed in this chapter concerning the fulfilment of technology convenience, culture-enriched, location-based, context-aware, and team-oriented mobile learning. It is economically, socially, politically, and technologically sustainable (Cisler, n.d.) since pervasive games are self-sustained by nature.

It is our hope that the game proposed in this study can be expanded into a wider area in the future to include more culture and history of Southern Taiwan into the instructional pervasive game, through which formal education materials can be brought more alive for the students.

## Acknowledgements

This research was supported by the National Science Council in Taiwan under projects NSC 98-2511-S-024-006-MY2, NSC 100-2628-S-024-002-MY3, NSC 100-2631-S-011-003.

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